

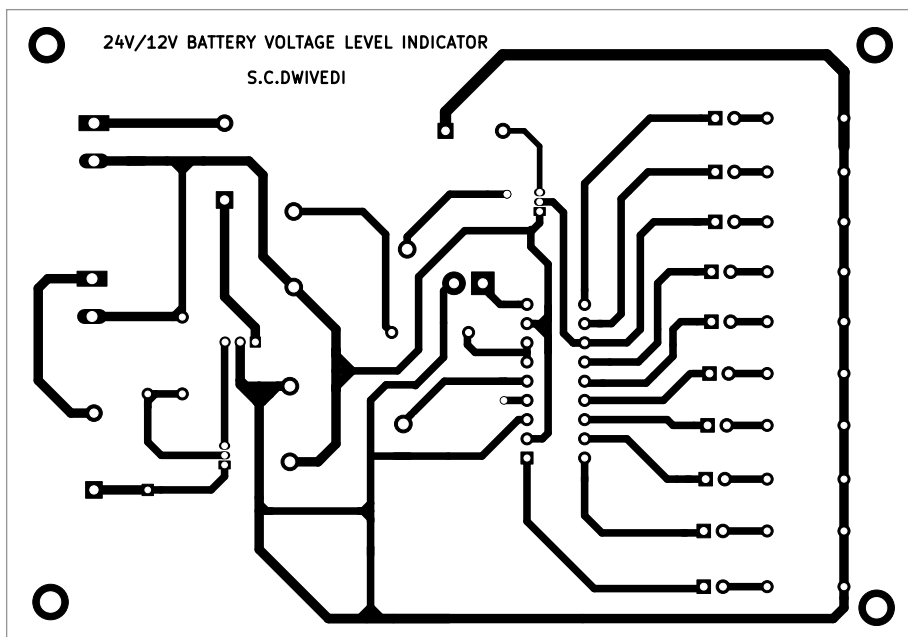


Fig. 4: Actual-size, single-side PCB layout

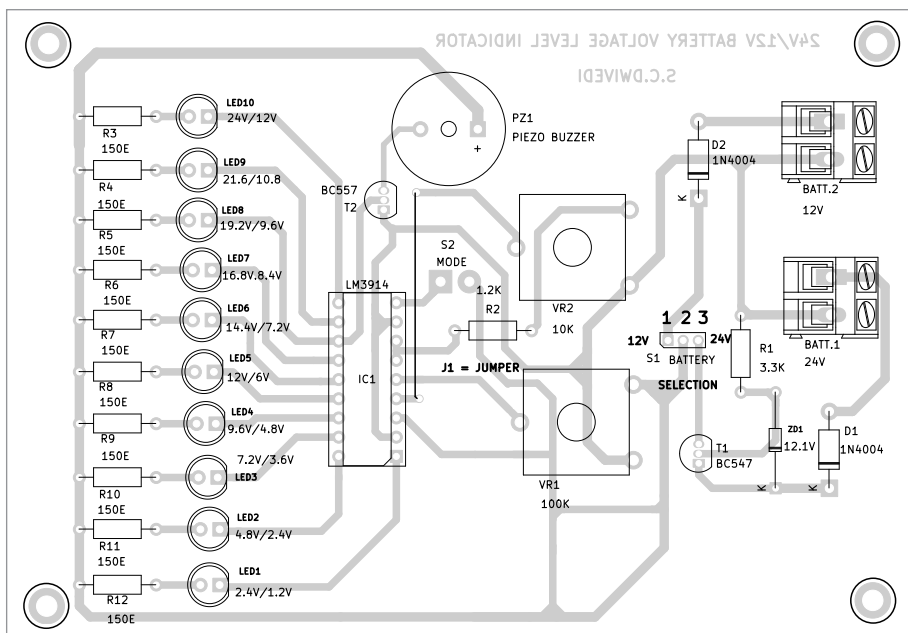


Fig. 5: Component layout of the PCB

black clip to its negative terminal. Flip switch S1 for checking a 24V battery's status. Position of switch S2 will depend on whether you want a dot display or a bar display.

Use presets VR1 and VR2 to calibrate the circuit. And you may use a 30V variable power supply instead of the 24V battery for calibration. Connect 12V from the variable pow-

er supply in place of 24V battery in the circuit. Adjust preset VR1 such that LED1 just starts glowing. Now increase the input DC voltage slowly to 24V and observe the LED's glowing status. The first LED (LED1) will start glowing at 2.4V and the second (LED2) at 4.8V, and so on. The last LED (LED10) will glow at 24V.

After this calibration the circuit

is ready to use. If you want to check the level of a 24V battery, connect it to the circuit using the crocodile clips and flip switch S1 towards 24V position.

To see the voltage level in dot mode, keep switch S2 open. If, say, LED9 glows, it means voltage level of the battery is around 90% of 24V, that is, around 21.6V. If LED10 glows, it means the battery is fully charged.

To see the battery voltage level in bar mode, flip switch S1 towards 24V position and turn switch S2 on. If, say, LED1 through LED9 start glowing, it means voltage level of the battery is around 90% of 24V, that is, around 21.6V. If all the ten LEDs (LED1 through LED10) glow, it means voltage level of the battery is full 24V.

The buzzer in circuit sounds when the battery's voltage level falls to 80% to indicate that the battery needs to be charged. The base of transistor T2 connected to pin12 of IC1 drives the piezo buzzer.

Construction and testing

An actual-size, single-side PCB for the 24V/12V battery status indicator is shown in Fig. 3 and its component layout in Fig. 4. After assembling the circuit on PCB, enclose it in a suitable plastic box. Switches S1 through S2 may be fixed on back side of the cabinet. The assembled PCB may be fixed inside the box's front side in such a way

that LED1 through LED10 are clearly visible.

BONUS: You can watch the VIDEO of the tutorial of this DIY project at: <https://www.electronicsforu.com/videos-slideshows/live-diy-battery-voltage-level-indicator> **EFY**

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